

AMENDMENT NO. 1 OCTOBER 2009
TO
IS 2811 : 1987 RECOMMENDATIONS FOR MANUAL TUNGSTEN INERT-GAS ARC
WELDING OF
AUSTENITIC STAINLESS STEEL

(First Revision)

(Page 2, clause 6) — Substitute the following for the existing clause:

6 ELECTRODES

6.1 Non-consumable electrodes made of tungsten (W) are used because of its high melting point. Any of the grades given in Table 3A can be used depending on the nature of the job.

Table 3A* Chemical Composition of Non-consumable Tungsten Electrodes

Grade	W Percent, <i>Min</i>	ThO ₂ Percent	ZrO ₂ Percent	CeO ₂ Percent	La ₂ O ₃ Percent	Other Oxides, Elements, Percent, <i>Max</i>
Tungsten (pure)	99.5	—	—	—	—	0.5
W (2% Ceria)	97.3	—	—	1.8-2.2	—	0.5
Tungsten (1% Lanthana)	98.3	—	—	—	0.8 - 1.2	0.5
Tungsten (1.5% Lanthana)	97.8	—	—	—	1.3 - 1.7	0.5
Tungsten (2% Lanthana)	97.3	—	—	—	1.8 - 2.2	0.5
Tungsten (1% Zirconia)	99.1	—	0.15- 0.40	—	—	0.5
Tungsten (1% Thoria)	98.3	0.8 - 1.2	—	—	—	0.5
Tungsten (2% Thoria)	97.3	1.7 - 2.2	—	—	—	0.5

NOTE — However, the following must be kept in mind while deciding to use tungsten electrodes containing Thorium Oxide as it is mildly radioactive. If welding is to be performed in confined spaces for prolonged periods of time or if electrode grinding dust might be ingested, special precautions for ventilation should be considered. As such, it is best to consider grades other than those containing thorium because of occupation health concerns. This standard discourages use of thoriated tungsten electrodes instead cerated tungsten electrodes recommended.

*Quantities indicated are in weight, percent.

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